

Program and Branch Name:
 BTech CSE - F

PBL Progress Report – II

Field	Details
Course Name	Computer Architecture & Parallel Processing
Course Code	CCSE0304
Faculty Name	Dr. Amrita Bhatnagar
Assignment Type	Group Assignment
Student(s) Name	Utkarsh Raj Shukla Vivek Kumar Vishal Gupta Yash Srivastava Nishant Kumar Mahto
ERP ID (I Year) / Roll No. (II Year onwards)	2501330100398 2501330100414 2501330100411 2501330100422 0261DCS009
Team/Group No. (<i>Group/Capstone only</i>)	Group No. - 91
Problem Statement / Project Title	Pharmacy Inventory Computing System
SDG(s)	SDG 3 – Good Health & Well-Being
Reporting Month	Month 2
Date of Submission	08/10/2026

Student Declaration

I/We certify that this assignment is my/our own work, with all sources appropriately acknowledged. I/We understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature(s):

S. No.	Student Name	Roll No.	Signature
1	Utkarsh Raj Shukla	2501330100398	Utkarsh Raj Shukla
2	Vivek Kumar	2501330100414	Vivek
3	Vishal Gupta	2501330100411	Vishal
4	Yash Srivastava	2501330100422	Yash
5	Nishant Kumar Mahto	0261DCS009	Nishant Kumar Mahto

Date: 08/10/2026

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Review 2: Development & Application

Review Coverage: Group Assignment — Unit 2 (remaining: division and logic operations, floating point arithmetic, ALU design, IEEE 754) and Unit 3 (Control Unit: instruction types, formats, instruction cycle and sub-cycles, micro-operations, program control, RISC, CISC, pipelining, hardwired and microprogrammed control, horizontal and vertical microprogramming, Flynn’s classification), built on the Unit 1 and Unit 2 study from Review 1.

Field	Details	
1. Previous Review Feedback	No feedback has been given by the faculty on Report I yet, so the team moved from concept study to implementation as planned in Report I.	
2. Action Taken on Previous Feedback	- Completed every action planned in Report I: finalised the requirements and features, prepared the flowchart and design, built the data/system structure, applied the Unit 3 concepts, and started implementation. - Went beyond the plan: the system is now a working web app, hosted on GitHub Pages, with a separate lab page for each CAPP unit for the review.	
3. Research / Data Collection Progress	Dataset: 10 sample medicines (analgesic, antibiotic, insulin, vaccine, etc.) with batch, units, pack size, reorder level, unit price, expiry, supplier, and Rx/cold-storage flags. Data file: assets/js/store.js in the repository. Study: Dr. Amrita Bhatnagar’s class notes and PPTs for Units 2 and 3; M. Morris Mano, <i>Computer System Architecture</i>; W. Stallings, <i>Computer Organization & Architecture</i>; C. Hamacher, <i>Computer Organization</i>. Standards / papers: IEEE Std 754-2019 (floating point arithmetic); Booth (1951), A signed binary multiplication technique, <i>Q. J. Mech. Appl. Math.</i> 4(2) – Booth’s algorithm; Flynn (1966), Very high-speed computing systems, <i>Proc. IEEE</i> 54(12) – Flynn’s classification.	
4. Unit/Module Concepts Applied	Concept	How it is applied
	Division (U2)	Restoring and non-restoring division turn loose tablets into packs (quotient) and loose tablets (remainder), e.g. $157 \div 10 = 15 \text{ strips} + 7$.
	Logic operations (U2)	An 8-bit status register per medicine (Low, Out, Expiring, Expired, Rx, Cold, High value). OR sets bits; an AND mask raises alerts and filters the list.
	ALU design (U2)	8-bit ALU with S2 S1 S0 select and C/Z/N/V flags. Sale = SUB, purchase = ADD; a borrow (C = 0) blocks overselling.
	IEEE 754, FP arithmetic (U2)	Prices shown as sign/exponent/fraction bits (32 and 64 bit). Stock value is computed in float32. Add/sub/mul/div are shown as align → operate → normalise → round → pack.
	Instruction types, formats (U3)	“PharmaCPU”: 16-bit words, 16 instructions, 5 formats (R, I, M, X, B); data-transfer, arithmetic, logical and program-control types.
	Instruction cycle, micro-ops (U3)	Fetch (T0–T2), decode (T3) and execute (T4...) run clock by clock as RTL micro-operations, e.g. $MAR \leftarrow PC$, $MBR \leftarrow M[MAR]$.
	Program control, RISC/CISC (U3)	JN/JZ/JNZ loops scan inventory records in memory. A 4-instruction RISC load/store sequence is compared with one CISC memory-to-memory instruction.
	Pipelining (U3)	5-stage IF–ID–EX–MEM–WB timing diagram with RAW hazards, operand forwarding and branch flushes.
	Hardwired vs micro-programmed (U3)	Hardwired: T-states × opcode decoder ($D_i \cdot T_j$). Microprogrammed: control memory, CAR and mapping logic; horizontal (24-bit) vs vertical (26-bit) microcode.
	Flynn’s classification (U3)	Animated SISD vs SIMD expiry scan (4 lanes → 4× fewer steps), with MISD and MIMD pharmacy examples.
5. Methodology / Approach Followed	Inventory: add, edit or sell a medicine → ALU ADD/SUB with flags → status register rebuilt with OR → alerts from AND masks → packs from division. CPU lab: assembly program → two-pass assembler → machine code → clock-by-clock fetch/decode/execute on inventory data at 0x100 → dynamic trace → pipeline timing. Tools: JavaScript, HTML/CSS, Node.js (tests), Git/GitHub, GitHub Actions, GitHub Pages.	

Field	Details				
6. Work Completed During Month 2	- Inventory module (CRUD, sale/purchase through the ALU, bit-mask filter, dashboard alerts, JSON import/export). - ALU Lab (ALU, division, Booth, logic ops, IEEE 754, FP arithmetic); Control Unit Lab (ISA, assembler, CPU, control unit views, pipeline, RISC/CISC, Flynn); automated tests; deployment.				
7. Analysis / Development / Implementation	Memory map: N at 0x100, then 3-word records [qty, reorder, status]; results at 0x1F0–0x1F2. Four assembly programs: low-stock count, total units, expiry scan (AND mask 0000 1100), record a sale. Example: selling 30 of 120 units runs SUB 120, 30 → 90 with Z=0, N=0, C=1. Selling 50 of 36 gives C = 0 (borrow), so the sale is refused. FP example: ₹6.40 × 36 in float32 → significands multiplied, normalised, rounded to nearest even → 230.40000915... (the bit-exact hardware result). Low-stock program: found Amoxicillin, Metformin and ORS (3 items), the same as the dashboard. It ran 85 instructions in 535 clock cycles (T-states).				
8. Results / Interim Outcomes	Program (sample data)		Instructions	Non-pipelined	Pipelined, no fwd
	Low-stock count		85	425 cycles	147 (2.89×)
	Total units in stock		55	275 cycles	97 (2.84×)
	Expiry alert scan		71	355 cycles	123 (2.89×)
- Forwarding saves 10 cycles per program; the remaining stalls are load-use hazards and taken branches. - FP results matched the hardware FPU in 80,000/80,000 random tests; all 84,374 automated checks pass.					
9. Evidence / Supporting Material	- Repository: github.com/utk042/CAPP-PBL - Live app: utk042.github.io/CAPP-PBL/ · Class notes and PPTs (Units 2 and 3) · Reference textbooks · Report I				
10. Challenges & Corrective Actions	- Non-restoring division failed for large divisors → widened the A register to n + 2 bits; all 8-bit cases verified. - Booth overflowed at −128 → limited inputs to ±127; every product in range verified. - FP rounding errors → exact significands with guard/round/sticky bits, compared with hardware. - Simulators too dense on phones → responsive layout with scrollable tables.				
11. Individual / Team Contribution	Member		Roll No.	Contribution	
	Utkarsh Raj Shukla		2501330100398	PharmaCPU ISA, assembler, instruction-cycle simulator, integration, GitHub	
	Vivek Kumar		2501330100414	ALU design, IEEE 754 encoding, floating point arithmetic	
	Vishal Gupta		2501330100411	Restoring/non-restoring division, Booth, logic ops and status register	
	Yash Srivastava		2501330100422	Pipelining, RISC vs CISC, Flynn’s classification, testing	
	Nishant Kumar Mahto		0261DCS009	Inventory UI and dashboard, control-unit views, report compilation	
12. Overall Progress (%)	50%				
13. Plan for Final Review	- Unit 4: cache simulator (direct and set-associative mapping) for medicine records, virtual memory, and MSI/MESI/Dragon coherence for several billing counters. - Unit 5: directory protocols, memory consistency, LL-SC and barrier synchronisation for concurrent sales, and interconnects for multi-branch stores; final report and demo.				

Faculty Evaluation:

- To be completed by Faculty in Moodle